

Subject Description Form

Subject Code	COMP2411				
Subject Title	Database Systems				
Credit Value	3				
Level	2				
Pre-requisite / Co-requisite / Exclusion	Pre-requisite: COMP1011/COMP1012/ENG2002				
Objectives	The objectives of this subject are to: - design, develop, implement, and administrate a database system of considerable complexity; and - possess enough background to evaluate various DBMSs of different data models and make the appropriate selection for an organisation.				
Intended Learning Outcomes	Upon completion of the subject, students will be able to: <u>Professional/academic knowledge and skills</u> (a) understand database management systems, as well as to apply the associated tools and techniques; (b) understand the principles and practices of database design and analysis; (c) identify the direction of database technology and their implication, and plan database developments; <u>Attributes for all-roundedness</u> (d) understand development of database technologies, e.g., web databases; and (e) achieve team outcomes, and develop presentation and technical writing skills.				
Subject Synopsis/ Indicative Syllabus	<table border="1"> <tr> <td>Topic</td></tr> <tr> <td> 1. Basic Concepts of Database System Database and its applications; DBMS design objectives and its components; three-level architecture; data independence. </td></tr> <tr> <td> 2. Database Design Entity-relationship model; functional dependencies; normalisation. </td></tr> <tr> <td> 3. Relational Data Model Relational structure; relational languages: relational algebra, relational calculus, SQL; relational constraints: entity constraints, referential integrity constraints and foreign keys. </td></tr> </table>	Topic	1. Basic Concepts of Database System Database and its applications; DBMS design objectives and its components; three-level architecture; data independence.	2. Database Design Entity-relationship model; functional dependencies; normalisation.	3. Relational Data Model Relational structure; relational languages: relational algebra, relational calculus, SQL; relational constraints: entity constraints, referential integrity constraints and foreign keys.
Topic					
1. Basic Concepts of Database System Database and its applications; DBMS design objectives and its components; three-level architecture; data independence.					
2. Database Design Entity-relationship model; functional dependencies; normalisation.					
3. Relational Data Model Relational structure; relational languages: relational algebra, relational calculus, SQL; relational constraints: entity constraints, referential integrity constraints and foreign keys.					

	4. File Structures and Physical Database Design File organisation; indexing and hashing.						
	5. Application Design and Query Processing Relational view definition and management; equivalence of query expressions, estimation of query-processing cost, join strategies; embedded SQL.						
	6. Implementation Issues Buffer management; transaction processing; concurrency control; crash and recovery; security and integrity.						
Teaching/ Learning Methodology	This subject emphasises the technical/practical aspects of database design and development. It is intended to equip the student with knowledge and practical experience on the real-life/industrial database application development. The lectures will be used to deliver course material that will be practiced/reinforced during the labs and tutorials.						
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed				
			a	b	c	d	e
	Continuous Assessment	30%					
	1. Assignments, Tests and Projects		✓	✓	✓	✓	✓
	Examination	70%	✓	✓	✓		
	Total	100%					
Student Study Effort Expected	Class contact:						
	▪ Lectures					39 Hrs.	
	▪ Tutorials/Lab					13 Hrs.	
	Other student study effort:						
	▪ Assignments, Tests, Projects, Exams					67 Hrs.	
	Total student study effort					119 Hrs.	
Reading List and References	Reference Books: 1. Elmasri, Ramez and Navathe, Shamkant B., <i>Fundamentals of database systems</i> , 7 th Edition, Pearson, 2016.						

	2. Kroenke, David M., Auer, David J., Vandenberg, Scott L. and Yoder, Robert C., <i>Database Processing: Fundamentals, Design, and Implementation</i>, 16th Edition, Pearson, 2021. 3. Mannino, Michael, <i>Database Design, Application Development, and Administration</i>, 7th Edition, SAGE Publications, Inc, 2023. 4. Silberschatz, Avi, Korth, Henry F. and Sudarshan, S., <i>Database System Concepts</i>, 7th Edition, McGraw Hill, 2019. https://db-book.com
--	---